Project Design Phase-I
Solution Architecture

Date	28 October 2026
Team ID	
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

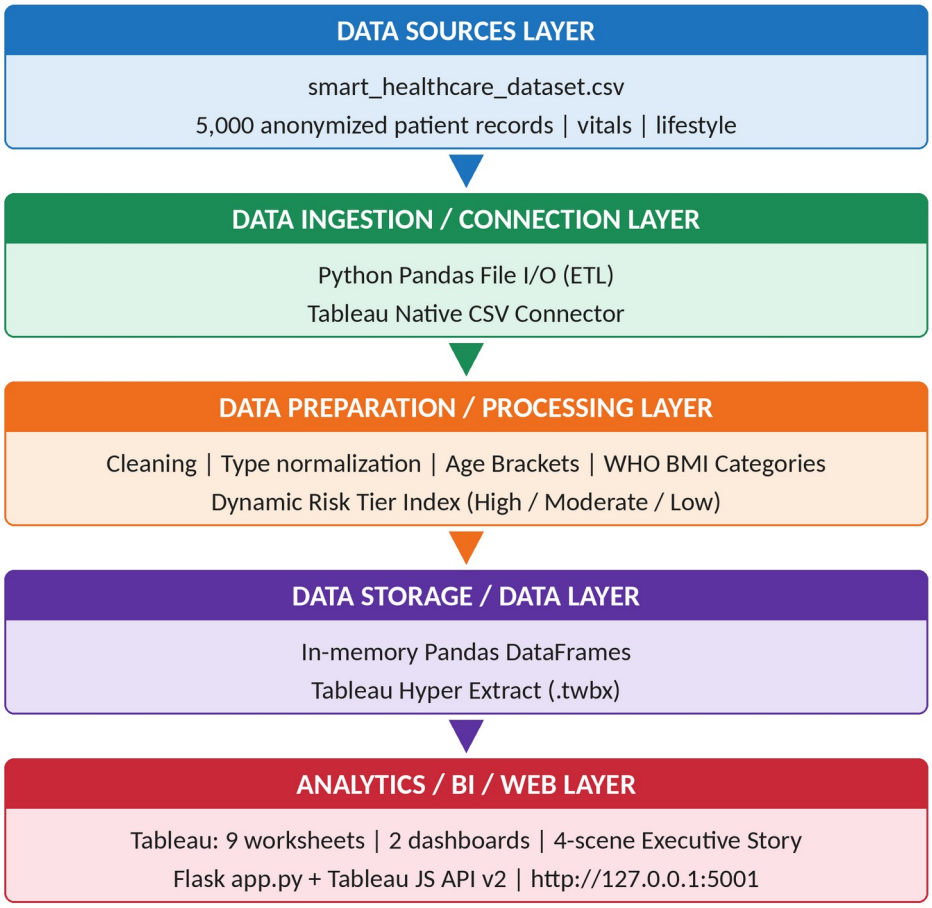
Solution Context:

Health360 bridges raw patient health data and clinical decision-making by creating an end-to-end analytics workflow. It processes 5,000 patient records across clinical vitals and lifestyle factors to deliver dynamic risk stratification, interactive Tableau dashboards, and a lightweight web deployment.

Architecture Goals:

- Ingest and clean raw CSV clinical data.
- Perform feature engineering (Age Brackets, WHO BMI Categories, Dynamic Risk Tier Index).
- Build a robust 5-tier architecture layer (Data Sources → Ingestion → Processing → Storage → Analytics/Web Layer).
- Deliver actionable insights over a web-embedded dashboard hosted on Flask.

Sample Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Sources Layer	Primary source: smart_healthcare_dataset.csv, 5,000 anonymized patient records containing vitals, demographic data, and lifestyle indicators.	CSV flat file
Data Ingestion / Connection Layer	Connectors: Python Pandas File I/O for ETL preprocessing; Tableau Native CSV Connector for Hyper extract building.	Python Pandas, Tableau CSV Connector
Data Preparation / Processing Layer	Data type normalization, handling missing values, calculating Age Brackets (18–34 to 65+), WHO BMI Categories, and the multi-variable Dynamic Risk Tier Index (High, Moderate, Low Risk).	Python Pandas, Tableau Calculated Fields

Data Storage / Data Layer	In-memory Pandas DataFrames for web service execution and a Tableau Hyper data extract (packaged in the .twbx workbook) for high-speed BI visual rendering.	Pandas DataFrames, Tableau Hyper Extract (.twbx)
Analytics / BI / Web Layer	Tableau rendering 9 worksheets, 2 master dashboards, and a 4-scene Executive Story; embedded in a Python Flask web application (app.py) via Tableau JS API v2 and served at http://127.0.0.1:5001.	Tableau Desktop / Public, Python Flask, Tableau JS API v2

Figure 1: Health360 End-to-End System Solution Architecture

